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Roll No

EE/EX-4003-CBGS

B.E. IV Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Digital Electronics Logic Design

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Convert the following:
 - i) $(0.828125)_{10} = (?)_2$
 - ii) $(18.6)_9 = (?)_{11}$
 - iii) $(234)_{10} = (?)_8$
 - iv) $(AF.16C)_{16} = (?)_8$
 - v) $(4163)_{10} + (7520)_8 = (?)_2$
- b) Explain error detecting and correcting codes with examples.
2. a) Implement $f(x, y, z) = \sum m(0, 3, 4, 5, 7)$ in NAND logic.
- b) Design a 32101 multiplexer using 8 to 1 multiplexers having an active LOW ENABLE input and a 2-to-4 decoder.
3. a) What do you understand by Universal Gate. Design AND, OR and NOT gates using NOR gate only.
- b) What do you understand by Multiplexer? Implement the following function using 8*1 multiplexer
 $F(A, B, C, D) = \sum(0, 1, 3, 4, 8, 9, 15)$

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PTO

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4.
 - a) What is Latch? What are the difference between Latch and Flip Flop? Explain D latch with truth table and logic diagram.
 - b) Explain J and K Flip Flop with truth table and logic diagram.
5.
 - a) Define the following and explain:
 - i) State Diagram
 - ii) State Table
 - iii) State Equation
 - iv) State Assignment
 - b) Explain the working of Johnson counter with suitable diagram.
6.
 - a) What do you understand by Counter? Give the difference between Synchronous and Asynchronous counter.
 - b) What is Shift Register? What are different types of shift registers? Explain.
7.
 - a) Explain the working of BCD counter.
 - b) Discuss the principle working of programmable array logic.
8. Write short notes on :
 - a) Encoder and Decoder
 - b) PLD
